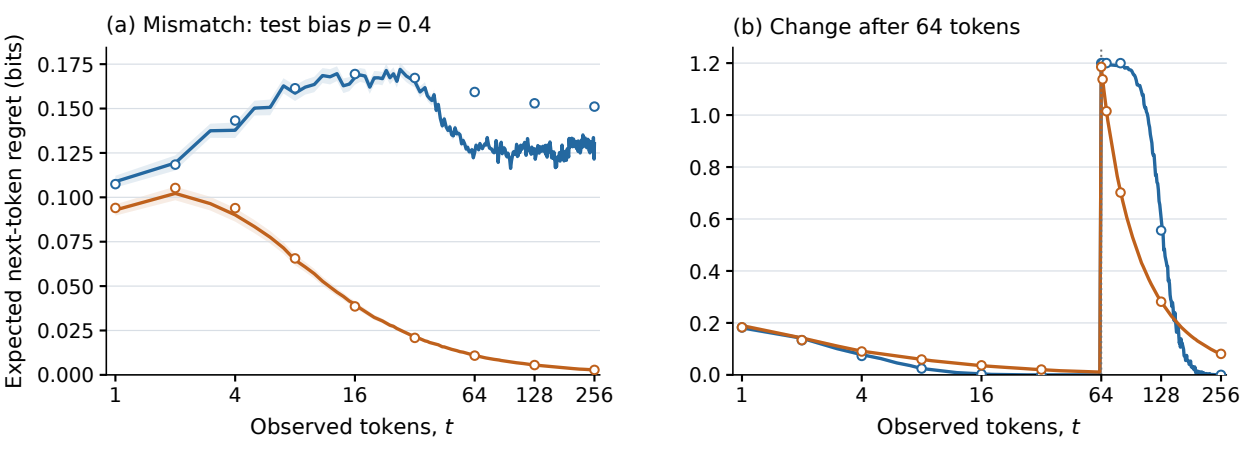
(a) Mismatch: test bias $p = 0.4$
Expected next-token regret (bits)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1
4
16
64
128
256
Observed tokens, t
(b) Change after 64 tokens
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
4
16
64
128
256
Observed tokens, t
Prior $\{0.2, 0.8\}$
Prior uniform on $[0, 1]$
Transformer
Informed Bayes (exact)